# Classification Metrics — The Senior Engineer's Study Guide

A complete reference and drill set for classical ML classification metrics. Built to close the gap that surfaced in your assessment: knowing what each metric measures, when to use it, how to read metric movements diagnostically, and how these fit into the broader ML stack.

## 1. Where These Metrics Sit in the Stack

An important framing point up front: **classification metrics are classical ML fundamentals, not deep learning specifics.** They apply to every supervised classifier — logistic regression, random forests, gradient boosting, neural networks, even LLMs used as classifiers. Getting this mental map right matters because senior interviewers test whether you know which concept lives at which layer.

| Layer | What lives here | Examples |
|---|---|---|
| Math / Stats foundations | Probability, linear algebra, optimization | Bayes rule, gradients, MLE |
| Classical ML | Supervised learning framework, metrics, model families, feature engineering | Confusion matrix, ROC-AUC, PR-AUC, F1, precision, recall — **this guide** |
| Deep learning | Neural network training specifics | Backprop, activations, batch norm, optimizers |
| LLMs / GenAI | Language-model-specific concerns | Tokenization, attention, RAG, RLHF, evals |
| MLOps | Production concerns spanning all of the above | Drift, monitoring, CI/CD for models |

> **Senior-level takeaway:** When you evaluate a churn model, a fraud model, an image classifier, or an LLM-based intent classifier — you use these same metrics. The metrics don't change; only the model does.

## 2. The Foundation — The Confusion Matrix

Every classification metric in existence is built from four numbers. If these aren't automatic, nothing downstream will click.

| | Actually Positive | Actually Negative |
|---|---|---|
| **Predicted Positive** | True Positive (TP)<br>Correct catch | False Positive (FP)<br>False alarm |
| **Predicted Negative** | False Negative (FN)<br>Missed catch | True Negative (TN)<br>Correctly left alone |

### Memorize by meaning, not by letter

- **TP**: You said churn, they churned. Model wins.
- **FP**: You said churn, they stayed. False alarm — you wasted a retention offer.
- **FN**: You said stay, they churned. Missed — you lost the customer.
- **TN**: You said stay, they stayed. Correctly left alone.

### Which error hurts more depends on the business

- **Churn**: FN usually costs more (losing a customer > wasting a $10 discount).
- **Cancer screening**: FN is catastrophic (missed diagnosis).
- **Spam filter**: FP is worse (real email lost).
- **Fraud detection**: depends on transaction size and friction cost of blocking a legit user.

**Rule:** Senior engineers always frame metric choice around **which error type is more expensive**, in dollars or user experience. If you can't say that in the first two sentences of your answer, you sound junior.

## 3. The Point Metrics

These are computed at **one specific decision threshold** (default 0.5, but always tunable).

### Accuracy

**Formula:** (TP + TN) / (TP + TN + FP + FN)

**Meaning:** Fraction of all predictions that were correct.

**The trap:** Useless on imbalanced data. If 95% of customers don't churn, a model that always predicts 'no churn' scores 95% accuracy while being completely worthless.

> **Rule:** Never quote accuracy alone on imbalanced problems. Interviewers bait you with this.

### Precision

**Formula:** TP / (TP + FP)

**Meaning:** Of everyone I flagged as positive, what fraction really was positive?

**Read as:** 'When my model says churn, how often is it right?'

**Business translation:** Trustworthiness of alarms. High precision = few false alarms.

**Optimize when:** False positives are costly (spam filter, wrongful fraud block).

### Recall (Sensitivity, True Positive Rate)

**Formula:** TP / (TP + FN)

**Meaning:** Of all real positives that exist, what fraction did I catch?

**Read as:** 'Of everyone who actually churned, how many did I flag?'

**Business translation:** Coverage. High recall = few misses.

**Optimize when:** False negatives are costly (cancer, fraud, churn where the customer is high value).

### The Precision-Recall Tradeoff (critical intuition)

You control this with the **threshold**. Lower the threshold → more 'positive' predictions → catch more real positives (recall ↑) but also more false alarms (precision ↓). Raise the threshold → the opposite. You cannot maximize both at once. Senior engineers always think in terms of this tradeoff, not a single 'best' number.

### F1 Score

**Formula:** 2 × (Precision × Recall) / (Precision + Recall)

**Meaning:** Harmonic mean of precision and recall.

**Why harmonic mean?** Arithmetic mean rewards being amazing at one metric even if terrible at the other. Harmonic mean punishes imbalance — if either metric is near zero, F1 is near zero. Forces both to be decent.

**Threshold-dependent:** F1 is measured at one specific threshold. Sensitive to where you draw the decision line.

**F-beta variants:** F2 weights recall higher, F0.5 weights precision higher. Use these when one error type is more costly than the other.

## Specificity (True Negative Rate)

**Formula:** TN / (TN + FP)

**Meaning:** Of all real negatives, what fraction did I correctly identify as negative?

**Where you see it:** Medical testing (a specific test rarely calls a healthy person sick). Also the complement of the ROC curve's x-axis (FPR = 1 – Specificity).

# 4. The Curve Metrics

These sweep the threshold from 0 to 1 and summarize performance across all possible operating points. They are **threshold-independent**.

## ROC Curve and ROC-AUC

**Axes:** False Positive Rate (x) vs True Positive Rate / Recall (y).

**FPR:** FP / (FP + TN).

**Interpretation:** As you lower the threshold, you catch more positives (TPR ↑) but also more false alarms (FPR ↑). The curve traces that tradeoff.

**AUC:** Area under the curve. Probability that the model scores a random positive higher than a random negative. Pure ranking quality.

- 0.5 = random guessing
- 0.7 = mediocre
- 0.8 = good
- 0.9+ = excellent
- 1.0 = perfect

**Critical weakness:** ROC-AUC is **inflated on imbalanced data**. The FPR denominator (TN + FP) is huge when negatives dominate, so false positives barely move the FPR. A model can look impressive by ROC-AUC while making terrible positive-class predictions in absolute terms.

## PR Curve and PR-AUC

**Axes:** Recall (x) vs Precision (y).

**Interpretation:** Tradeoff between 'catching them all' and 'being right when you flag them'.

**Baseline is different from ROC:** Random guessing on ROC gives AUC = 0.5. Random guessing on PR gives AUC ≈ base rate of the positive class. So if only 5% of customers churn, a random model gets PR-AUC ≈ 0.05.

**Why it dominates for imbalanced problems:** It ignores true negatives entirely. It only cares about how well you handle the minority class. It cannot be flattered by the sea of easy negatives.

**Senior-level rule:** On imbalanced data, trust PR-AUC over ROC-AUC. Interviewers test this constantly.

# 5. The Comparison Table (memorize this)

| Metric | Threshold-dependent? | Handles imbalance well? | What it measures |
|---|---|---|---|
| Accuracy | Yes | No — misleading | Overall correctness |

| Metric | Threshold-dependent? | Handles imbalance well? | What it measures |
|---|---|---|---|
| Precision | Yes | Yes | Trustworthiness of positive predictions |
| Recall | Yes | Yes | Coverage of actual positives |
| F1 | Yes | Yes | Balance of precision and recall |
| ROC-AUC | No | No — can be inflated | Overall ranking quality |
| PR-AUC | No | Yes — best for imbalance | Positive-class ranking quality |

# 6. Reading Metric Movements Diagnostically

This is the skill your assessment was really testing. Given a set of metric changes, you should be able to reason backward to the cause. This is what separates senior engineers from mid-level candidates.

| Pattern | Likely Cause | Where to look |
|---|---|---|
| ROC-AUC steady, PR-AUC drops | Base rate shift (label drift) — proportion of positives changed | Compare positive rate in training vs recent production |
| Both AUCs steady, F1 drops | Score calibration shift — ranking OK, scores shifted | Recalibrate; retune threshold; check probability histograms |
| Everything drops together | Genuine feature drift or concept drift — model no longer understands the data | Compare feature distributions; check delayed labels for concept drift |
| Recall drops, precision rises (or reverse) | Threshold is now in the wrong place | Sweep threshold; find new operating point |
| Precision drops on positives, else fine | New subpopulation the model doesn't recognize | Segment analysis; find which slice is misclassified |

## Applied to your assessment scenario

**Offline:** ROC-AUC 0.91, PR-AUC 0.62, F1 0.58. **Live:** ROC-AUC 0.73, PR-AUC 0.21, F1 0.19.

Relative drops: ROC-AUC –20%, PR-AUC –66%, F1 –67%.

PR-AUC and F1 collapsed harder than ROC-AUC. That asymmetry points strongly to a **base rate shift and/or calibration drift** as a leading hypothesis. The model still has ranking signal (ROC-AUC is degraded but not destroyed), but its positive-class predictions have fallen apart. Classic signature of label drift and/or feature drift with score miscalibration.

# 7. Senior-Level Context You Also Need

## Calibration

A model can have great AUC but poorly calibrated probabilities. If your model outputs 0.8 for a customer, does that customer actually churn 80% of the time? Study Brier score, reliability diagrams, Platt scaling, isotonic regression. Calibration matters when you use scores for downstream decisions (expected value calculations, cost-weighted decisions).

## Multi-class extensions

- **Macro-average:** unweighted mean across classes. Treats each class equally. Best when you care about minority classes.
- **Micro-average:** weighted by class frequency. Dominated by the majority class. Reflects overall accuracy.
- **Weighted-average:** mean weighted by class support. A compromise.

## Business-metric alignment

Every technical metric should map to a business outcome. Cost-weighted evaluation: assign a dollar cost to FP and FN, minimize expected cost. This separates senior engineers from mid-level — the ability to say 'F1 is not the right metric here because a false negative costs us $500 in lost revenue while a false positive costs us $10 in a retention offer.'

## Class imbalance handling

- **Threshold tuning** — cheapest, most effective first move.
- **Class weights** — clean, works for most model families.
- **SMOTE / synthetic oversampling** — creates synthetic minority examples. Watch for leakage across CV folds.
- **Downsampling majority class** — fast, but breaks calibration; correct scores afterward if needed.
- **Focal loss** — deep learning trick that down-weights easy examples. Useful for extreme imbalance.

## Statistical significance

Bootstrap confidence intervals on AUC. Is that 0.91 → 0.73 drop statistically significant or noise from a small validation sample? Real senior engineers ask this and know how to answer.

# 8. Worked Examples

## Example 1: Compute all point metrics from a confusion matrix

**Setup:** Churn model, 1,000 customers, threshold 0.5.

| | Actually Churn (100) | Actually Stay (900) |
|---|---|---|
| Predicted Churn | TP = 70 | FP = 90 |
| Predicted Stay | FN = 30 | TN = 810 |

**Precision** = 70 / (70 + 90) = 70/160 = **0.4375**. Of the 160 customers we flagged, only 44% actually churned — a lot of false alarms.

**Recall** = 70 / (70 + 30) = 70/100 = **0.70**. We caught 70 of the 100 real churners.

**F1** = 2 × (0.4375 × 0.70) / (0.4375 + 0.70) = **0.538**. Reflects that precision is dragging us down.

**Accuracy** = (70 + 810) / 1000 = **0.88**. Notice how misleading this looks compared to the actually mediocre F1.

**Specificity** = 810 / (810 + 90) = **0.90**. High, because there are lots of easy negatives.

## Example 2: The base-rate flip — why PR-AUC crashes when ROC-AUC barely moves

Suppose training has a 10% churn base rate and production shifts to a 3% base rate (customers got stickier because a competitor collapsed). The model's ranking ability is unchanged. But now:

- At the deployed threshold, most flagged customers are false positives (because there just aren't as many real churners around).
- Precision plummets: TP shrinks, FP stays roughly the same.
- Recall may hold or drop.
- F1 collapses.
- PR-AUC collapses (its baseline dropped from 0.10 to 0.03).
- ROC-AUC barely moves — the ranking is fine.

**This is exactly the shape of the numbers in your assessment.**

## Example 3: Threshold tuning

Your model outputs probabilities. At threshold 0.5 you get F1 = 0.40. You sweep thresholds:

| Threshold | Precision | Recall | F1 |
|---|---|---|---|
| 0.3 | 0.30 | 0.90 | 0.45 |
| 0.4 | 0.38 | 0.80 | 0.51 |

| Threshold | Precision | Recall | F1 |
|---|---|---|---|
| 0.5 | 0.50 | 0.60 | 0.55 |
| 0.6 | 0.65 | 0.45 | 0.53 |
| 0.7 | 0.80 | 0.25 | 0.38 |

F1 peaks around threshold 0.5. But if false negatives cost 10× more than false positives, the **cost-optimal** threshold is around 0.4 — not 0.5, and not the F1-optimal one.

# 9. Practice Problems

Work each problem before checking the answer key. Write out your reasoning before computing. That reasoning is what senior interviewers are grading.

## Tier 1 — Fundamentals

***Problem 1***

A fraud model classifies 10,000 transactions. It predicts 400 as fraud, of which 300 really are fraud. There were 500 true fraud cases in total. Compute precision, recall, F1, and accuracy. Which metric is most misleading here and why?

***Problem 2***

A model has ROC-AUC = 0.95 on a dataset with 99.5% negatives. Should you trust the model? What single additional metric would you compute before deciding, and why?

***Problem 3***

A colleague says 'we should use F1 because it balances precision and recall.' In a medical screening context where missed diagnoses can kill patients but false positives just mean a follow-up test, what would you push back on and what metric would you propose instead?

## Tier 2 — Reasoning

***Problem 4***

Two models: A has ROC-AUC 0.88 and PR-AUC 0.42. B has ROC-AUC 0.85 and PR-AUC 0.55. Which would you deploy for a highly imbalanced churn problem and why? What if the classes were roughly balanced?

***Problem 5***

You retrained a model. On the same test set, ROC-AUC stayed at 0.90, but F1 dropped from 0.55 to 0.42. Precision fell, recall rose. What is the most likely explanation and what single action would you take?

***Problem 6***

Your model's PR-AUC is 0.30 and F1 is 0.35. Random guessing on this dataset would yield PR-AUC ≈ 0.05 (positive class is 5% of the data). Is the model good or bad? Frame your answer in terms of lift over baseline and the business context you would need to know.

## Tier 3 — Senior-level scenarios

***Problem 7 (diagnostic)***

An image classifier for defective parts on a production line had validation Precision 0.92, Recall 0.88, F1 0.90. In production after 3 weeks, Precision is 0.94, Recall is 0.55, F1 is 0.69. The camera hardware, model artifact, and preprocessing haven't changed. What hypotheses would you form, in ranked order, and what specific check would you run for each?

***Problem 8 (cost-weighted)***

You have a churn model whose F1-optimal threshold is 0.5, giving precision 0.60 and recall 0.70. False negatives (losing a customer) cost the business $200 on average. False positives (sending an unnecessary retention offer) cost $15. The base rate is 8% and you score 100,000 customers per month. Should you use the F1-optimal threshold? If not, in which direction should you shift it, and what would you compute to decide?

### *Problem 9 (calibration)*

Your model's ROC-AUC is 0.90 but the marketing team complains that customers with predicted churn probability of 0.8 are only churning at a 45% rate. What is going on, why does it matter for the business, and what techniques would you apply?

### *Problem 10 (multi-class)*

You have a 5-class classifier with the following per-class F1 scores: A 0.92, B 0.88, C 0.30, D 0.85, E 0.90. Class supports are: A 4000, B 3000, C 100, D 2500, E 400. Compute macro-F1 and weighted-F1 by hand (approximately). Which one better reflects the model's real usefulness and why? What would you do about class C?

### *Problem 11 (integration — the assessment-style question)*

A recommendation model for a streaming service was showing offline NDCG of 0.68 and offline hit-rate@10 of 0.55. Two weeks after deployment, online engagement is down 22%, but offline metrics on the last 7 days of production data still show NDCG 0.65 and hit-rate@10 of 0.52. The model, preprocessing, and inference API are unchanged. Walk through your investigation. Cover: (a) why offline and online metrics disagree, (b) whether this is drift or something else, (c) three specific hypotheses ranked by likelihood, (d) what monitoring you would add. This mirrors the structure of your assessment question — practice writing this out end-to-end.

## 10. Answer Key

Read the reasoning, not just the numbers. If your reasoning matched but the numbers were off, you're in good shape — just drill arithmetic. If the reasoning was off, re-read the relevant metric section.

### Problem 1

TP = 300, FP = 100, FN = 200, TN = 9,400.

- Precision = 300/400 = **0.75**
- Recall = 300/500 = **0.60**
- F1 = 2 × 0.75 × 0.60 / (0.75 + 0.60) = **0.667**
- Accuracy = (300 + 9400) / 10000 = **0.97**

**Accuracy is most misleading.** It reads 97% but the model misses 40% of fraud. On this 5% base rate dataset, a model that always predicts 'not fraud' would score 95% accuracy while catching nothing. Precision, recall, and PR-AUC tell the real story.

### Problem 2

Do **not** trust the model based on ROC-AUC alone. On a 0.5% positive class, ROC-AUC is notoriously inflated because the enormous negative pool makes FPR insensitive to false positives. Compute **PR-AUC** next. If PR-AUC is 0.05 (baseline for 0.5% positive class) then the model is worthless. If it's 0.60, the model actually works.

### Problem 3

Push back: F1 gives equal weight to precision and recall, but in medical screening they are not equally important. A missed diagnosis can be fatal; a false positive triggers a follow-up test. Propose **F2 or recall-at-fixed-precision**, or ideally a **cost-weighted expected value**. Frame it as: 'What's the cost of each error type?' rather than 'What's a balanced metric?'

### Problem 4

For imbalanced churn: deploy **B**. Its higher PR-AUC means better positive-class handling, which is what matters when positives are rare. B's slightly lower ROC-AUC is likely just reflecting the well-known ROC-AUC inflation effect on imbalanced data. For balanced classes, the two metrics roughly agree, so **A** (higher ROC-AUC) would be the choice. This question tests whether you understand *context-dependent metric selection.*

### Problem 5

ROC-AUC unchanged means the model's **ranking is identical**. What changed is where the threshold sits relative to the score distribution — likely the new model's scores shifted (perhaps more calibrated toward the positive class). This is a **threshold problem, not a model problem**. **Action:** sweep thresholds on the validation set and pick the F1-optimal one. Don't retrain.

### Problem 6

PR-AUC 0.30 vs baseline 0.05 is a **6× lift over random**. This is meaningful signal. Whether it's 'good' depends entirely on business context: what's the cost of a missed positive, and what's the operational cost of acting on the model? A 6× lift model can be extremely valuable if positives are expensive to miss and interventions are cheap. Senior answer: **never call a metric 'good' or 'bad' without a baseline and a business context.**

## Problem 7

Recall crashed while precision rose. This means the model's still confident when it fires, but it's **firing less often on real defects**. Ranked hypotheses:

- **1. Concept drift on the positive class.** New defect types appeared that the model wasn't trained on. Check: sample recent misses, have engineers label them, see if they look novel.
- **2. Feature drift on positives.** Lighting, camera position, or upstream process changed. Check: compare distributions of scored features on positives across time windows.
- **3. Score calibration shift.** Same defects but scored lower now. Check: histogram of prediction scores over time; look for a leftward shift.
- **4. Threshold now wrong for the operating regime.** Check: PR curve on recent labeled data.

Note that the deployed threshold could be temporarily lowered as a short-term fix while you diagnose. Long-term: retrain or fine-tune on the new defect samples.

## Problem 8

Expected cost per customer at threshold 0.5: with 8% base rate, of 100k customers, 8,000 real positives and 92,000 negatives. At precision 0.60 recall 0.70: TP = 5,600, FN = 2,400, FP = 3,733. Cost = 2,400 × \$200 + 3,733 × \$15 = \$480,000 + \$56,000 = **\$536,000**.

Because FN is 13× more expensive than FP, you should **lower the threshold** to catch more churners even at the cost of more false positives. Compute expected cost across a range of thresholds and pick the minimum — almost certainly well below 0.5. This is **cost-weighted threshold selection** and it is the senior-level answer to nearly every 'what threshold?' question.

## Problem 9

The model's **ranking is good (high AUC)** but its **probabilities are miscalibrated**. It's telling marketing that customers have an 80% chance of churning when they really have a 45% chance. This matters because marketing likely uses the probability directly for expected-value calculations ('spend up to \$80 to save an 80%-risk customer'), and the miscalibration causes systematic overspending.

**Techniques:** Platt scaling (fit a logistic regression on model scores → true labels), isotonic regression (nonparametric, more flexible but needs more data), or temperature scaling for neural nets. Evaluate with a **reliability diagram** and **Brier score** before and after.

## Problem 10

Total support = 10,000. Weighted-F1 ≈ (0.92 × 4000 + 0.88 × 3000 + 0.30 × 100 + 0.85 × 2500 + 0.90 × 400) / 10000 ≈ **0.881**. Macro-F1 = average of the five = (0.92 + 0.88 + 0.30 + 0.85 + 0.90) / 5 = **0.77**.

Weighted-F1 hides the disaster on class C because it has only 100 support. Macro-F1 exposes it by treating each class equally. **Which is right depends on business:** if class C matters (e.g., it's a rare but important category like 'fraud' or 'critical fault'), macro-F1 is the honest metric. If C is a rare-and-unimportant category, weighted-F1 is fine.

**What to do about C:** gather more examples (targeted labeling, oversampling), use class weights, or consider a hierarchical model. Check whether the misclassifications are being confused with a specific other class — that often points to a labeling ambiguity you can fix upstream.

## Problem 11

Full answer sketch — write this out yourself and compare:

- **(a) Why offline and online metrics disagree:** Offline eval on production data with delayed feedback typically measures ranking against historical clicks. Online engagement measures actual user behavior including exposure effects, feedback loops, novelty, and position bias. A model can rank existing items well while still underperforming because it recommends items users don't actually engage with in the real UI.
- **(b) Drift or something else:** Offline metrics on recent data are near baseline, which argues against pure data drift. This suggests **selection bias in offline evaluation** (you're only evaluating on items that had feedback, which the model may have caused), or **a feedback-loop / exposure-bias issue** where the deployed model has shifted the distribution of what users see.
- **(c) Three hypotheses, ranked:** (1) Feedback loop — deployed model narrowed the recommendation set, hurting exploration. (2) Position/exposure bias not corrected offline — offline eval overstates quality. (3) Concept drift on user preferences that offline eval on old-preference data misses.
- **(d) Monitoring:** Online A/B test metrics (CTR, session length, retention), diversity and coverage of recommendations, exposure distribution, engagement by recency of item, and counterfactual offline evaluation using inverse propensity scoring.

## 11. Study Roadmap

### Immediate (this week)

- Draw the confusion matrix from memory daily until automatic.
- For each metric, articulate: 'goes up when X, down when Y, use it when Z.'
- Redo problems 1-6 without notes.

### Short-term (2-3 weeks)

- Master the diagnostic table in Section 6. Given any metric movement, form 2-3 ranked hypotheses.
- Study calibration: Platt scaling, isotonic regression, reliability diagrams, Brier score.
- Study class imbalance handling: thresholding, class weights, SMOTE, downsampling, focal loss.
- Redo problems 7-11 and write them out end-to-end as if for an interviewer.

### Medium-term (1-2 months)

- Move to MLOps: drift detection (feature drift, label drift, concept drift — three distinct things), monitoring architectures, training-serving skew, shadow deployment, champion-challenger.
- Multi-class and ranking metrics: macro/micro/weighted averages, NDCG, MRR, MAP.
- Cost-sensitive learning and cost-weighted evaluation.
- Practice full assessment-style scenarios like Problem 11 weekly.

### The mental habit to build

Before touching a model, ask three questions: **(1) What is the base rate? (2) What does a false positive cost? (3) What does a false negative cost?** Every metric choice and every threshold decision flows from these three answers. Senior engineers ask these first; junior engineers ask them last, if at all.